

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

ANS: "No Bootable Device" → Faulty SSD/HDD, bad cable, or wrong boot order.

"CMOS Checksum Error" → Dead CR2032 motherboard battery.

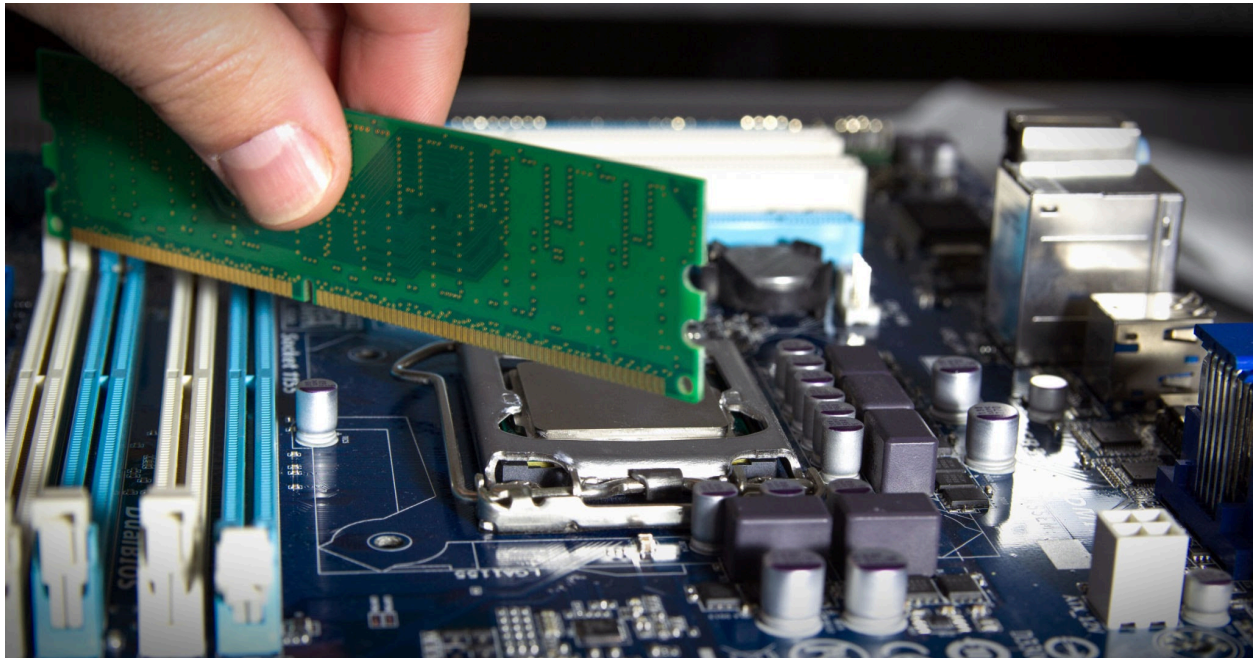
"CPU Fan Error" → Fan unplugged, dead, or not spinning fast enough.

"Memory Test Failed" (Beep Code) → Loose or faulty RAM.

"SMART Error / Disk Failure Imminent" → Failing SSD/HDD (back up data immediately).

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

ANS:



Troubleshooting:

1. **Reseat RAM:** Remove memory sticks, wipe contacts, and snap back into place.
2. **Test One Stick:** Boot with a single RAM module at a time to isolate a bad stick.

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

ANS:

- BIOS / Firmware Vendor: American Megatrends, Inc. (AMI) — Aptio Setup Utility
- BIOS Version: 2.17.1245 (Project Version: 37.01.04.00)
- Motherboard / System Developer: Oracle / Sun Microsystems (Sun Server platform running AMI Aptio core firmware)

How to find these details on your local PC or VM:

1. Accessing BIOS: Restart your PC and repeatedly press Del, F2, or F12 (or F2 / Esc on a Virtual Machine) before Windows loads.
2. Finding Manufacturer & Version: Look at the top or bottom banner on the Main tab. Alternatively, press **Win + R** in Windows, run **msinfo32**, and check BaseBoard Manufacturer and BIOS Version/Date.



**You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.**

ANS:

A **CMOS checksum error** occurs when the checksum value generated during POST does not match the value saved in the CMOS RAM. This mismatch indicates that saved BIOS settings (date, time, boot order, RAM speed) were corrupted or completely wiped, usually due to a loss of auxiliary power when the PC was powered off. To protect the system, the BIOS automatically resets all settings to factory defaults.

```
Press <Del> to enter setup, <F12> Display Boot Menu.

BIOS Version: P01-A3
Release Date: 02/18/2011
CPU : Intel(R) Core(TM) i5-2500 CPU @ 3.30GHz
Stepping: 7  CPUID: 000206a7
Cache L1 Size: 256 KB
Cache L2 Size: 1024 KB
Cache L3 Size: 6144 KB
The IMC is operating with DDR3 1333
Dual Channel Mode
8192MB OK

USB Device(s) : 0 Keyboard, 2 Mice, 5 MASS, 2 HUBs
USB MASS #0:Generic-Compact Flash 1.00
USB MASS #1:Generic-SM/xD-Picture 1.00
USB MASS #2:Generic-SD/MMC 1.00
USB MASS #3:Generic-MS/MS-Pro/MS 1.00
USB MASS #4:Generic-SD/MMC/MS/MSPRO 1.00

Detected ATA/ATAPI Devices...
SATA Port0: WDC WD10EALS-00ZBA0
SATA Port1: ATAPI DVD A DH16ABSH
Error: CMOS checksum error - Defaults loaded
Press F1 to Continue. Press Del to Enter BIOS Setup.
```

How to Resolve It

1. **Replace the CMOS Battery:**
 - Power down the PC, unplug the power cable, and open the side panel.
 - Locate the **CR2032 3V coin-cell battery** on the motherboard.
 - Press the release latch, remove the dead battery, and insert a fresh CR2032 battery.
2. **Reconfigure and Save BIOS Settings:**
 - Turn on the PC and press **Del** or **F2** to enter BIOS/UEFI Setup.
 - Correct the **Date and Time**.
 - Re-enable custom options like XMP/EXPO, AHCI mode, or specific boot priority orders.
 - Press **F10** to save changes and exit.

DAY 4

Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

ANS:

Steps Taken:

1. **Power Off:** Shut down PC, unplug power cable, and hold power button 10 seconds to drain residual charge.
2. **Ground & Open:** Remove side panel and touch chassis metal to discharge static.
3. **Release:** Push down slot retention clips to eject the RAM stick.
4. **Remove:** Lift module out by its edges.
5. **Re-insert:** Align stick notch with slot key and press down firmly on both ends until clips click locked.

Observed Issues:

- **High Resistance:** Required significant downward force to seat the stick fully.
- **Partial Latching:** If pushed unevenly, one side fails to lock, risking a loose connection and no-POST error.

Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

ANS:

1. **Identify Symptoms:** Note artifacts, screen tearing, driver crashes under load, or no display.
2. **Check Cables & Ports:** Plug cable directly into GPU (not motherboard) and test with a different HDMI/DisplayPort cable.
3. **Inspect Power & Reseat:** Reseat card in PCIe slot, verify latch clicks, and ensure PCIe power cables are flush.
4. **Reinstall Drivers (DDU):** Boot into Safe Mode, clear old drivers using Display Driver Uninstaller (DDU), and install fresh WHQL drivers.
5. **Stress Test & Cross-Test:** Run FurMark for 20 minutes to test stability under load, or test the card in a second PC.

Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

ANS:

Before Reconnecting (Drive Unplugged)

- **POST Error Message:** "Reboot and Select proper Boot device" or "No Bootable Device Found".
- **BIOS Indicators:**
 - Under **Storage/SATA/NVMe Configuration**, the port shows "**Not Present**" or "**Empty**".
 - Under **Boot Priority**, the drive model name disappears, leaving only network/PXE boot options.



After Reconnecting (Drive Plugged In)

- **POST Behavior:** POST passes without errors and boots straight into the operating system.
- **BIOS Indicators:**
 - Storage port updates to display the drive's exact model and capacity (e.g., *Samsung SSD 980 1TB*).
 - Drive reappears in **Boot Priority List** as Option #1.

Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

ANS:

Component First: RAM (System Memory)

Reasoning: Corrupted RAM cells trigger instant kernel crashes and random reboots. GPU issues usually crash only during gaming; storage issues freeze the OS rather than cause sudden restarts.

RAM Diagnostic Steps:

1. **Disable XMP/EXPO:** Reset RAM to stock JEDEC speeds in BIOS to eliminate unstable overclocks.
2. **Run MemTest86:** Boot from a MemTest86 USB drive and run 4 passes to scan for hardware errors.
3. **Isolate Sticks:** Remove all memory modules and test each stick individually in slot 2/4 to find the bad module.

DAY 5

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

ANS:

Check Charger & Outlet: Verify adapter LED and laptop charging light turn on to confirm wall power reaches the device.

Perform Static Drain: Unplug power and hold power button 20 seconds to clear trapped residual motherboard charge.

Isolate Battery: Disconnect internal battery and boot on charger alone to detect a shorted battery blocking startup.

Inspect DC/USB-C Port: Check for bent pins or loose connections preventing power transmission.

Strip Peripherals & RAM: Remove external USBs and secondary RAM/SSDs to eliminate components triggering Over-Current Protection.

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.
Hint: Consider checking both hardware connections and basic display settings.

ANS:

Verify Monitor & Input: Check if monitor power LED is on and manually select the correct input channel (HDMI/DisplayPort).

Check Cable & GPU Port: Ensure the cable is plugged into the **discrete GPU** (lower horizontal slot), not the motherboard. Unplug and re-seat both ends.

Reset Display Driver: Press **Win + Ctrl + Shift + B** on the keyboard to force Windows to restart the graphics driver.

Reseat RAM & GPU: Power off, open case, and firmly re-seat memory sticks and graphics card until slot latches click.

Clear CMOS: Remove CR2032 motherboard battery for 5 minutes to reset BIOS settings to default, then test with 1 RAM stick.

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Test Mouse (Hardware): Plug dongle/mouse into a second PC. If non-responsive, battery is dead or mouse is faulty.

Test USB Port: Move dongle to a different USB port on the same PC. If it works, original port is dead or loose.

Test Settings (Driver): Use keyboard (**Win + X** → **Device Manager**) to check for yellow warning icons under *Mice* or *USB Controllers*.

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

***Hint:** Think about fans, dust, and thermal paste.*

ANS:

Dead or Unplugged Fans: Check if CPU cooler and case fans spin on boot. Listen for bearing noise or feel for airflow out the back.

Dust Clogging: Open the chassis and inspect the heatsink fins, mesh filters, and GPU fans for dense dust buildup blocking heat dissipation.

Dried Thermal Paste: Remove the CPU cooler to check if thermal compound is dried, cracked, or unevenly covering the processor IHS.